

>>> Topological invariant of 4-manifolds based on a 3-group

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>>> A sketch of the talk

► 3-group and 3-gauge theory

→ based on *R. Picken and J. Faria Martins*, *Diff. Geom. Appl.* 29, 179 (2011), [arXiv:0907.2566](#).

► $3BF$ action

→ Models with relevant dynamics *T. Radenković and M. Vojinović*, *J. High Energy Phys.* 10, 222 (2019), [arXiv:1904.07566](#).

► Quantization of the topological $3BF$ theory

→ the state sum Z is an example of Porter's TQFT for $d = 4$ and $n = 3$
T. Porter, *J. Lond. Math. Soc.* (2)58, No. 3, 723 (1998), MR 1678163.

► Pachner move invariance

→ The construction of the state sum Z and a proof that the $3BF$ state sum is invariant under Pachner moves.

T. Radenković and M. Vojinović, [arXiv: 2201.02572](#).

→ This is a generalization of the state sum based on the classical $2BF$ action with the underlying 2-group structure

F. Girelli, H. Pfeiffer and E. M. Popescu, *Jour. Math. Phys.* 49, 032503 (2008), [arXiv:0708.3051](#).

► Conclusions

>>> 3-groups

2-crossed module $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{_, _\}_P)$

- * Groups G , H , and L ;
- * maps ∂ and δ ($\partial\delta = 1_G$);
- * an action $\triangleright$ of the group G on all three groups;
- * a map $\{_, _\}_P$ called the *Peiffer lifting*:

$$\{_, _\}_P : H \times H \rightarrow L.$$

Certain axioms hold true among all these maps:

1. $\delta(\{h_1, h_2\}_P) = \langle h_1, h_2 \rangle_P, \quad \forall h_1, h_2 \in H,$
2. $[l_1, l_2] = \{\delta(l_1), \delta(l_2)\}_P, \quad \forall l_1, l_2 \in L.$ Here, the notation $[l, k] = lkl^{-1}k^{-1}$ is used;
3. $\{h_1 h_2, h_3\}_P = \{h_1, h_2 h_3 h_2^{-1}\}_P \partial(h_1) \triangleright \{h_2, h_3\}_P, \quad \forall h_1, h_2, h_3 \in H;$
4. $\{h_1, h_2 h_3\}_P = \{h_1, h_2\}_P \{h_1, h_3\}_P \{\langle h_1, h_3 \rangle_P^{-1}, \partial(h_1) \triangleright h_2\}_P, \quad \forall h_1, h_2, h_3 \in H;$
5. $\{\delta(l), h\}_P \{h, \delta(l)\}_P = l(\partial(h) \triangleright l^{-1}), \quad \forall h \in H, \quad \forall l \in L.$

>>> The **3BF** theory

One can now generalize the notion of parallel transport from curves to surfaces and volumes.

- * Given a 2-crossed module, one can define a 3-connection, an ordered triple (α, β, γ) , where α , β , and γ are algebra-valued differential forms,

$$\begin{aligned}\alpha &= \alpha^\alpha{}_\mu \tau_\alpha dx^\mu, & \alpha &\in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{g}), \\ \beta &= \beta^a{}_{\mu\nu} t_a dx^\mu \wedge dx^\nu, & \beta &\in \mathcal{A}^2(\mathcal{M}_4, \mathfrak{h}), \\ \gamma &= \gamma^A{}_{\mu\nu\rho} T_A dx^\mu \wedge dx^\nu \wedge dx^\rho, & \gamma &\in \mathcal{A}^3(\mathcal{M}_4, \mathfrak{l}).\end{aligned}\tag{1}$$

- * Then introduce the line, surface and volume holonomies,

$$g = \mathcal{P}\exp \int_\gamma \alpha, \quad h = \mathcal{P}\exp \int_S \beta, \quad l = \mathcal{P}\exp \int_V \gamma.\tag{2}$$

- * The corresponding fake 3-curvature $(\mathcal{F}, \mathcal{G}, \mathcal{H})$ is defined as:

$$\begin{aligned}\mathcal{F} &= d\alpha + \alpha \wedge \alpha - \partial\beta, & \mathcal{G} &= d\beta + \alpha \wedge^\triangleright \beta - \delta\gamma, \\ \mathcal{H} &= d\gamma + \alpha \wedge^\triangleright \gamma + \{\beta \wedge \beta\}_{\text{pf}}.\end{aligned}\tag{3}$$

>>> The $3BF$ theory

At this point one can construct the so-called $3BF$ theory.

- * For a manifold $\mathcal{M}_4$ and the 2-crossed module $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{_, _\}_{\text{pf}})$, that gives rise to 3-curvature $(\mathcal{F}, \mathcal{G}, \mathcal{H})$, one defines the $3BF$ action as

$$S_{3BF} = \int_{\mathcal{M}_4} \langle B \wedge \mathcal{F} \rangle_{\mathfrak{g}} + \langle C \wedge \mathcal{G} \rangle_{\mathfrak{h}} + \langle D \wedge \mathcal{H} \rangle_{\mathfrak{l}}. \quad (4)$$

- * $3BF$ theory is a topological gauge theory,
- * it is based on the 3-group structure,
- * it is a generalization of an ordinary BF theory for a given Lie group G .
- * Physically relevant models

The constrained $2BF$ actions for

- * *Yang-Mills field*,
- * and *Einstein-Cartan gravity*,

and constrained $3BF$ actions describing

- * *Klein-Gordon field*,
- * *Dirac field*,
- * *Weyl fields*,
- * and *Majorana fields*

coupled gravity in the standard way are formulated.

>>> 3-gauge theory

- * Curves are labeled with the elements of G , and the elements are composed as

$$\bullet \xleftarrow{g_1} \bullet \xleftarrow{g_2} \bullet = \bullet \xleftarrow{g_1 g_2} \bullet$$

- * Surfaces are labeled with the elements $h \in H$. We split the boundary into two curves, the source curve $g_1 \in G$ and the target curve $g_2 \in G$,

$$\bullet \begin{array}{c} \xleftarrow{g_1} \\ \Downarrow h \\ \xleftarrow{g_2} \end{array} \bullet ,$$

so that the surface $h \in H$ satisfies:

$$\partial(h) = g_2 g_1^{-1} .$$

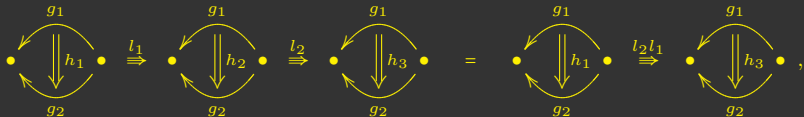
- * Volumes are labeled with the elements $l \in L$. We split the boundary into the source surface $\partial_3^-(l) = h_1$ and the target surface $\partial_3^+(l) = h_2$, and the common boundary of h_1 and h_2 we split into the source curve $\partial_2^-(l) = g_1$ and the target curve $\partial_2^+(l) = g_2$,

$$\bullet \begin{array}{c} \xleftarrow{g_1} \\ \Downarrow h_1 \\ \xleftarrow{g_2} \end{array} \bullet \quad \xRightarrow{l} \quad \bullet \begin{array}{c} \xleftarrow{g_1} \\ \Downarrow h_2 \\ \xleftarrow{g_2} \end{array} \bullet ,$$

$$\delta(l) = h_2 h_1^{-1} .$$

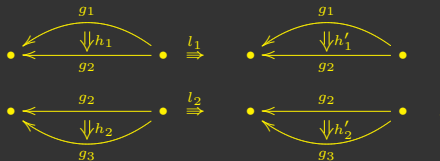
>>> 3-gauge theory

- * *Upward composition.* The upward composition of 3-morphisms (g_1, h_1, l_1) and (g_1, h_2, l_2) , when they are compatible, when $\partial_3^+(l_1) = \partial_3^-(l_2)$,



$$(g_1, h_2, l_2) \#_3 (g_1, h_1, l_1) = (g_1, h_1, l_2 l_1). \quad (5)$$

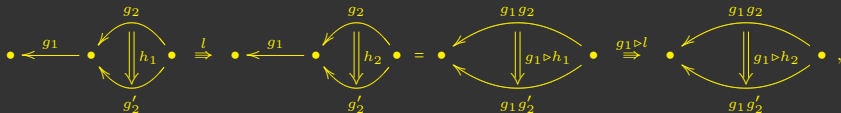
- * *Vertical composition.* The vertical composition of two 3-morphisms (g_1, h_1, l_1) and (g_2, h_2, l_2) , when they are compatible, when $\partial_2^+(l_1) = \partial_2^-(l_2)$,



$$(g_2, h_2, l_2) \#_2 (g_1, h_1, l_1) = (g_1, h_2 h_1, l_2 (h_2 \triangleright' l_1)). \quad (6)$$

>>> 3-gauge theory

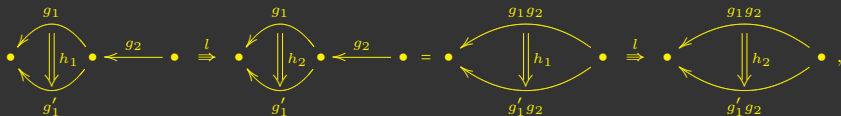
- * *Whiskering of the 3-morphisms with morphisms.* Whiskering of a 3-morphism by a morphism **from the left** is the composition of a volume $l \in L$ and curve $g_1 \in G$ from the left, when they are compatible, when $\partial_1^+(l) = \partial_1^-(g_1)$,



$$\bullet \xleftarrow{g_1} \bullet \begin{array}{c} \xrightarrow{g_2} \\ \Downarrow h_1 \\ \xleftarrow{g'_2} \end{array} \bullet \xrightarrow{l} \bullet \xleftarrow{g_1} \bullet \begin{array}{c} \xrightarrow{g_2} \\ \Downarrow h_2 \\ \xleftarrow{g'_2} \end{array} \bullet = \bullet \begin{array}{c} \xrightarrow{g_1 g_2} \\ \Downarrow g_1 \triangleright h_1 \\ \xleftarrow{g_1 g'_2} \end{array} \bullet \xrightarrow{g_1 \triangleright l} \bullet \begin{array}{c} \xrightarrow{g_1 g_2} \\ \Downarrow g_1 \triangleright h_2 \\ \xleftarrow{g_1 g'_2} \end{array} \bullet , \quad (7)$$

$$g_1 \#_1 (g_2, h_1, l) = (g_1 g_2, g_1 \triangleright h, g_1 \triangleright l) .$$

One can whisker a 3-morphism by a morphism **from the right**, when they are compatible, $\partial_1^-(l) = \partial_1^+(g_2)$,

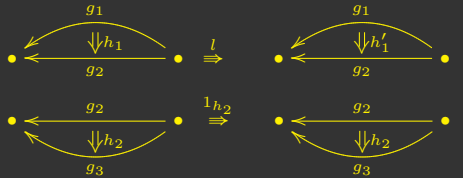


$$\bullet \begin{array}{c} \xrightarrow{g_1} \\ \Downarrow h_1 \\ \xleftarrow{g'_1} \end{array} \bullet \xleftarrow{g_2} \bullet \xrightarrow{l} \bullet \begin{array}{c} \xrightarrow{g_1} \\ \Downarrow h_2 \\ \xleftarrow{g'_1} \end{array} \bullet \xleftarrow{g_2} \bullet = \bullet \begin{array}{c} \xrightarrow{g_1 g_2} \\ \Downarrow h_1 \\ \xleftarrow{g'_1 g_2} \end{array} \bullet \xrightarrow{l} \bullet \begin{array}{c} \xrightarrow{g_1 g_2} \\ \Downarrow h_2 \\ \xleftarrow{g'_1 g_2} \end{array} \bullet , \quad (8)$$

$$(g_1, h_1, l) \#_1 g_2 = (g_1 g_2, h_1, l) .$$

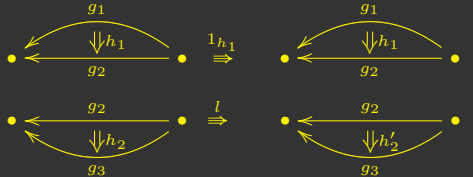
>>> 3-gauge theory

- * *Whiskering of 3-morphisms with 2-morphisms*. Whiskering of a 3-morphism with a 2-morphisms from below, when they are compatible, $\partial_2^+(l) = \partial_2^-(h_2)$,



$$(g_1, h_1, l) \#_2 (g_2, h_2) = (g_1, h_2 h_1, h_2 \triangleright' l). \quad (9)$$

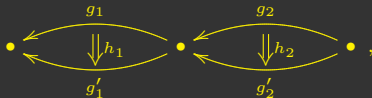
- * Whiskering a 3-morphism by 2-morphism from above, when they are compatible, when $\partial_2^-(l) = \partial_2^+(h_1)$,



$$(g_1, h_1) \#_2 (g_2, h_2, l) = (g_1, h_2 h_1, l). \quad (10)$$

>>> 3-gauge theory

- * *The interchanging 3-arrow.* The horizontal composition of two 2-morphisms h_1 and h_2 , when they are compatible, when $\partial_1^-(h_1) = \partial_1^+(h_2)$,



that results in a 3-morphism l , with source surface and target surfaces

$$\partial_3^-(l) = ((g_1, h_1) \#_1 g'_2) \#_2 (g_1 \#_1 (g_2, h_2)), \quad \partial_3^+(l) = (g'_1 \#_1 (g_2, h_2)) \#_2 ((g_1, h_1) \#_1 g_2).$$

One obtains,

$$(g_1, h_1) \#_1 (g_2, h_2) = (g_1 g_2, h_1 g_1 \triangleright h_2, \{h_1, g_1 \triangleright h_2\}_p^{-1}). \quad (11)$$

>>> 3-gauge theory

Lemma

Let us consider a triangle, (jkl) . The edges $(jk), j < k$, are labeled by group elements $g_{jk} \in G$ and the triangle $(jkl), j < k < l$, by element $h_{jkl} \in H$.

$$\begin{array}{c}
 \begin{array}{ccccc}
 & g_{kl} & & g_{jk} & \\
 & \swarrow & & \swarrow & \\
 l \bullet & & k \bullet & & j \bullet \\
 & \searrow & \Downarrow h_{jkl} & \searrow & \\
 & g_{jl} & & &
 \end{array}
 & = &
 \begin{array}{ccccc}
 & g_{kl} & & g_{jk} & \\
 & \swarrow & & \swarrow & \\
 l \bullet & & k \bullet & & j \bullet \\
 & \searrow & \Downarrow h_{jkl} & \searrow & \\
 & \partial(h_{jkl}) g_{kl} g_{jk} & & &
 \end{array}
 \end{array} \quad (12)$$

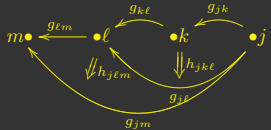
The curve $\gamma_1 = g_{kl}g_{jk}$ is the source and the curve $\gamma_2 = g_{jl}$ is the target of the surface morphism $\Sigma: \gamma_1 \rightarrow \gamma_2$, labeled by the group element h_{jkl} ,

$$g_{jl} = \partial(h_{jkl})g_{kl}g_{jk} . \quad (13)$$

>>> 3-gauge theory

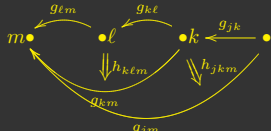
Lemma

Let us consider a tetrahedron, $(jklm)$.



$$= (g_{lm}g_{jl}, h_{jlm})\#_2(g_{lm}\#_1(g_{kl}g_{jk}, h_{jkl})) = (g_{lm}g_{kl}g_{jk}, h_{jlm}(g_{lm} \triangleright h_{jkl})).$$

(14)



$$= (g_{km}g_{jk}, h_{jkm})\#_2((g_{lm}g_{kl}, h_{klm})\#_1g_{jk}) = (g_{lm}g_{kl}g_{jk}, h_{jkm}h_{klm}).$$

(15)

Moving from surface shown on the diagram (14) to the surface shown on the diagram (15) is determined by the group element l_{jklm} ,

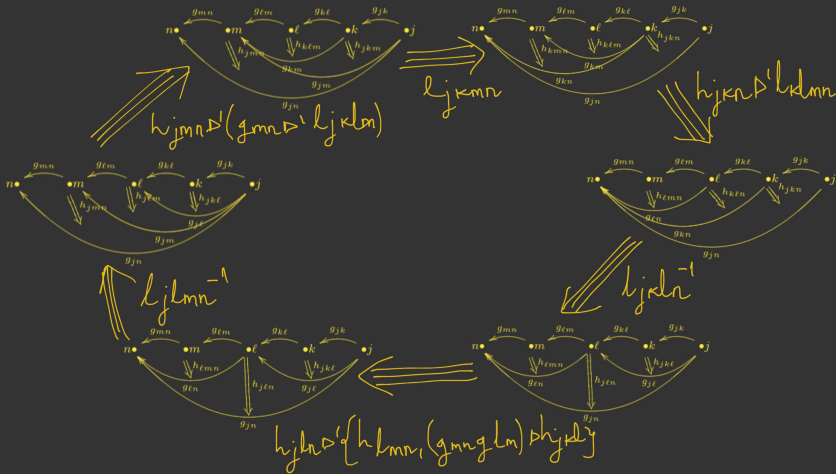
$$h_{jkm}h_{klm} = \delta(l_{jklm})h_{jlm}(g_{lm} \triangleright h_{jkl}).$$

(16)

>>> 3-gauge theory

Lemma (δ_L)

We consider a 4-simplex, $(jklmn)$. We cut the 4-simplex volume along the surface $h_{jmn}g_{mn} \triangleright (h_{jlm}g_{lm} \triangleright h_{jkl})$.



>>> 3-gauge theory

Lemma (δ_L)

After the upward composition of these 3-morphisms, the obtained 3-morphism is the identity morphism with source and target surface $\mathcal{V}_1 = \mathcal{V}_2 = h_{jmn}g_{mn} \triangleright (h_{j\ell m}g_{\ell m} \triangleright h_{jkl})$,

$$l_{j\ell mn}^{-1} h_{j\ell n} \triangleright' \{h_{\ell mn}, (g_{mn}g_{\ell m}) \triangleright h_{jkl}\}_P l_{jkl n}^{-1} (h_{jkn} \triangleright' l_{k\ell mn}) l_{jkmn} h_{jmn} \triangleright' (g_{mn} \triangleright l_{jklm}) = e. \quad (17)$$

>>> Quantization of the topological **3BF** theory

We want to construct a *state sum model* from the classical S_{3BF} action by the usual spinfoam quantization procedure.

$$Z = \int \mathcal{D}\alpha \mathcal{D}\beta \mathcal{D}\gamma \mathcal{D}B \mathcal{D}C \mathcal{D}D \exp \left(i \int_{M_4} \langle B \wedge \mathcal{F} \rangle_{\mathfrak{g}} + \langle C \wedge \mathcal{G} \rangle_{\mathfrak{h}} + \langle D \wedge \mathcal{H} \rangle_{\mathfrak{l}} \right). \quad (18)$$

↪ The formal integration over the Lagrange multipliers B , C , and D leads to:

$$Z = \mathcal{N} \int \mathcal{D}\alpha \mathcal{D}\beta \mathcal{D}\gamma \delta(\mathcal{F})\delta(\mathcal{G})\delta(\mathcal{H}). \quad (19)$$

↪ Discretization of the 3-connection:

- ▶ $\alpha \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{g}) \mapsto g_{\epsilon} \in G$ coloring the edges $\epsilon = (jk) \in \Lambda_1$,
- ▶ $\beta \in \mathcal{A}^2(\mathcal{M}_4, \mathfrak{h}) \mapsto h_{\Delta} \in H$ coloring the triangles $\Delta = (jkl) \in \Lambda_2$,
- ▶ $\gamma \in \mathcal{A}^3(\mathcal{M}_4, \mathfrak{l}) \mapsto l_{\tau} \in L$ coloring the tetrahedrons $\tau = (jklm) \in \Lambda_3$.

$$\left. \begin{array}{ll} \int \mathcal{D}\alpha & \mapsto \Pi_{(jk) \in \Lambda_1} \int_G dg_{jk} \\ \int \mathcal{D}\beta & \mapsto \Pi_{(jkl) \in \Lambda_2} \int_H dh_{jkl} \\ \int \mathcal{D}\gamma & \mapsto \Pi_{(jklm) \in \Lambda_3} \int_L dl_{jklm} \end{array} \right\} \longrightarrow \text{The discretization of path integral measures.}$$

>>> Quantization of the topological $3BF$ theory

↪ The condition $\delta(\mathcal{F})$ is discretized as

$$\delta(\mathcal{F}) = \prod_{(jk\ell) \in \Lambda_2} \delta_G(g_{jk\ell}), \quad \delta_G(g_{jk\ell}) = \delta_G(\partial(h_{jk\ell}) g_{k\ell} g_{jk} g_{j\ell}^{-1}). \quad (20)$$

↪ The condition $\delta(\mathcal{G})$ on the fake curvature 3-form reads

$$\delta(\mathcal{G}) = \prod_{(jk\ell m) \in \Lambda_3} \delta_H(h_{jk\ell m}), \quad (21)$$

$$\delta_H(h_{jk\ell m}) = \delta_H(\delta(l_{jk\ell m}) h_{j\ell m} (g_{\ell m} \triangleright h_{jk\ell}) h_{k\ell m}^{-1} h_{jk m}^{-1}). \quad (22)$$

↪ The condition $\delta(\mathcal{H})$ is discretized as

$$\delta(\mathcal{H}) = \prod_{(jk\ell mn) \in \Lambda_4} \delta_L(l_{jk\ell mn}), \quad (23)$$

$$\delta_L(l_{jk\ell mn}) = \delta_L(l_{j\ell mn}^{-1} h_{j\ell n} \triangleright' \{h_{\ell mn}, (g_{mn} g_{\ell m}) \triangleright h_{jk\ell}\}_{\mathbb{P}} l_{jk\ell n}^{-1} (h_{jkn} \triangleright' l_{k\ell mn}) l_{jk mn} h_{jmn} \triangleright' (g_{mn} \triangleright l_{jk\ell m})). \quad (24)$$

...all off this $\implies$

$$Z = \mathcal{N} \prod_{(jk) \in \Lambda_1} \int_G dg_{jk} \prod_{(jk\ell) \in \Lambda_2} \int_H dh_{jk\ell} \prod_{(jk\ell m) \in \Lambda_3} \int_L dl_{jk\ell m} \left(\prod_{(jk\ell) \in \Lambda_2} \delta_G(g_{jk\ell}) \right) \left(\prod_{(jk\ell m) \in \Lambda_3} \delta_H(h_{jk\ell m}) \right) \left(\prod_{(jk\ell mn) \in \Lambda_4} \delta_L(l_{jk\ell mn}) \right). \quad (25)$$

This expression can be made independent of the triangulation if one appropriately chooses the constant factor $\mathcal{N}$.

Definition

Let $\mathcal{M}_4$ be a compact and oriented combinatorial 4-manifold, and $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{_, _\}_{\text{pf}})$ be a 2-crossed module. The state sum of *topological higher gauge theory* is defined by

$$\begin{aligned}
 Z = & |G|^{-|\Lambda_0|+|\Lambda_1|-|\Lambda_2|} |H|^{|\Lambda_0|-|\Lambda_1|+|\Lambda_2|-|\Lambda_3|} |L|^{-|\Lambda_0|+|\Lambda_1|-|\Lambda_2|+|\Lambda_3|-|\Lambda_4|} \\
 & \times \left(\prod_{(jk) \in \Lambda_1} \int_G dg_{jk} \right) \left(\prod_{(jkl) \in \Lambda_2} \int_H dh_{jkl} \right) \left(\prod_{(jk\ell m) \in \Lambda_3} \int_L dl_{jk\ell m} \right) \\
 & \times \left(\prod_{(jkl) \in \Lambda_2} \delta_G \left(\partial(h_{jkl}) g_{kl} g_{jk} g_{jl}^{-1} \right) \right) \left(\prod_{(jk\ell m) \in \Lambda_3} \delta_H \left(\delta(l_{jk\ell m}) h_{j\ell m} (g_{\ell m} \triangleright h_{jk\ell}) h_{k\ell m}^{-1} h_{jkm}^{-1} \right) \right) \\
 & \times \left(\prod_{(jk\ell mn) \in \Lambda_4} \delta_L \left(l_{j\ell mn}^{-1} h_{j\ell n} \triangleright' \{h_{\ell mn}, (g_{mn} g_{\ell m}) \triangleright h_{jk\ell}\}_{\text{p}} l_{jk\ell n}^{-1} (h_{jkn} \triangleright' l_{k\ell mn}) l_{jkmn} h_{jmn} \triangleright' (g_{mn} \triangleright l_{jk\ell m}) \right) \right).
 \end{aligned}
 \tag{26}$$

Here $|\Lambda_0|$ denotes the number of vertices, $|\Lambda_1|$ edges, $|\Lambda_2|$ triangles, $|\Lambda_3|$ tetrahedrons, and $|\Lambda_4|$ 4-simplices of the triangulation.

>>> 1 ↔ 5 Pachner move



Right side

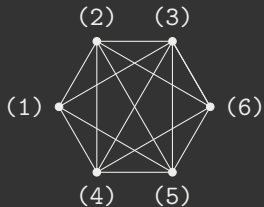
$$Z_{\text{right}}^{1 \leftrightarrow 5} = |G|^{-11} |H|^{-4} |L|^{-1} \int_{G^5} \prod_{(jk) \in M_1} dg_{jk} \int_{H^{10}} \prod_{(jkl) \in M_2} dh_{jkl} \int_{L^{10}} \prod_{(jklm) \in M_3} dl_{jklm} \cdot \left(\prod_{(jkl) \in M_2} \delta_G(g_{jkl}) \right) \left(\prod_{(jklm) \in M_3} \delta_H(h_{jklm}) \right) \left(\prod_{(jklmn) \in M_4} \delta_L(l_{jklmn}) \right) Z_{\text{remainder}}. \quad (27)$$

Left side

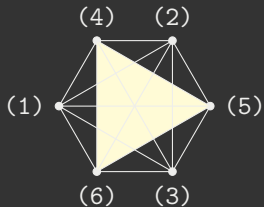
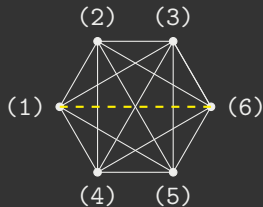
$$Z_{\text{left}}^{1 \leftrightarrow 5} = |G|^{-5} |H|^0 |L|^{-1} \delta_L(l_{23456}) Z_{\text{remainder}}. \quad (28)$$

The $Z_{\text{remainder}}$ denotes the part of the state sum that is the same on both sides of the move, and thus irrelevant for the proof of invariance.

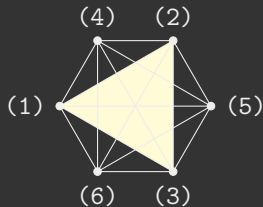
>>> 2 ↔ 4 and 3 ↔ 3 Pachner moves



2 ↔ 4



3 ↔ 3



>>> Synopsis

- * 2-crossed modules and 3-gauge theory
- * Physically relevant models -The constrained $2BF$ actions describing the *Yang-Mills field* and *Einstein-Cartan gravity*, and constrained $3BF$ actions describing the *Klein-Gordon*, *Dirac*, *Weyl* and *Majorana fields* coupled to Yang-Mills fields and gravity in the standard way.
- * Starting from the notion of Lie 3-groups, we generalize the integral picture of gauge theory to a 3-gauge theory that involves curves, surfaces, and volumes labeled with elements of non-Abelian groups.
- * The definition of the discrete state sum model of topological higher gauge theory in dimension $d=4$.
- * We prove that the state sum is well defined, i.e., invariant under the Pachner moves and thus independent of the chosen triangulation.

>>> Synopsis

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- * Physically relevant models -The constrained $2BF$ actions describing the *Yang-Mills field* and *Einstein-Cartan gravity*, and constrained $3BF$ actions describing the *Klein-Gordon*, *Dirac*, *Weyl* and *Majorana fields* coupled to Yang-Mills fields and gravity in the standard way.
- * Starting from the notion of Lie 3-groups, we generalize the integral picture of gauge theory to a 3-gauge theory that involves curves, surfaces, and volumes labeled with elements of non-Abelian groups.
- * The definition of the discrete state sum model of topological higher gauge theory in dimension $d=4$.
- * We prove that the state sum is well defined, i.e., invariant under the Pachner moves and thus independent of the chosen triangulation.

Thank you for your attention!